

LESSON 8.10

SOLVING SYSTEMS OF EQUATIONS BY GRAPHING FOR REAL-WORLD CONTEXTS



LESSON INTRODUCTION

MA.8.AR.4.3 Given a mathematical or real-world context, solve systems of two linear equations by graphing.

LEARNING TARGETS

- I can solve system of linear equations by graphing given a real-world context.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.

ESSENTIAL QUESTIONS

- How do you solve a real-world problem modeled with a system of linear equations by graphing?
- How do you interpret the solution(s) to a system of equations modeling a real-world context?

LESSON NARRATIVE

In this lesson, students apply the skills they have been working on over the last few lessons to solve problems involving systems of equations that model real-world contexts. The lesson focuses on students solving the systems of linear equations by graphing and then interpreting the solution within the context. Embedded in this lesson is the opportunity to review modeling and interpreting linear relationships from Unit 7.

COMPONENT SUMMARY

8.10.1 WARM-UP (~5 MIN.)

Notice and wonder with speed and distance graph

8.10.3 GUIDED PRACTICE (10-15 MIN.)

Graph and solve systems of linear equations in a real-world context

8.10.5 WRAP-UP (~5 MIN.)

Self-reflection of actions during the lesson

8.10.2 GUIDED INSTRUCTION (10-15 MIN.)

Graph and solve systems of linear equations in a real-world context

8.10.4 COLLABORATION (10-15 MIN.)

Graph and solve systems of linear equations in a real-world context

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- System of equations

MATERIALS

- (Optional) Graphing utility
- (Optional) Scientific calculator

ADDITIONAL PREPARATION

- 8.10.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly identify or interpret the slope and y-intercept in the real-world context.
- Students may approximate solutions incorrectly due to their lack of numeracy.
- Students may confuse the independent and dependent variables when interpreting the solution to systems of equations in context.

THINGS TO CONSIDER

- Consider ways to spiral content from Unit 7 as students are graphing and interpreting equations in this lesson.

LITERACY AND LANGUAGE ROUTINES

Three Reads

In 8.10.4 Collaboration, encourage students to use the “Three Reads” language routine to help comprehend the text. Have students read the text first for basic comprehension, second for analyzing the language and mathematical structure, and lastly to brainstorm possible solution paths.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students will be asked to make connections between systems of linear equations and real-world contexts. They will solve systems of linear equations by graphing and interpret their findings within the real-world context given. Throughout the lesson, students must graph and solve a system of linear equations prior to relating their solution back to the context.

DIFFERENTIATE INSTRUCTION

For students who struggle working with real-world contexts, consider providing students with highlighters or colored pencils to mark up the text as they read and digest the information. This will lead to better comprehension of the text and also illuminate which pieces of information they are drawn to, miss, or show confusion about. Offering students time to briefly discuss each context and what they understand from it with their partner before graphing and solving the problems will be helpful for helping the students to better comprehend the contexts and for the teacher to hear if there are any misconceptions that need to be addressed.

Additional differentiation opportunities are denoted throughout the Lesson Guide.

8.10.1 WARM-UP: NOTICE AND WONDER

DIFFERENTIATE INSTRUCTION

The video is short. Continue playing it more than once to support students as they answer question 1b, as they may want to collect additional information from the video.

TEACHER MOVES

Students' graphs will vary. If the students see the video, it shows three different versions of the race:

- Full speed
- Rich Eisen getting a 10-yard head start
- Julio Jones at half speed with Rich Eisen at full speed

Students also have the option to sketch what they think the graph will look like from the image if they are unable to view the video.

Depending on which they have in mind to sketch the graph, their graphs may look different.

If time allows, ask a few students to share their graphs and either explain which option they had in mind, OR allowing the class to guess which option they used based on their graph.



8.10.1 Warm-Up: Notice and Wonder

1. Julio Jones was drafted to the Atlanta Falcons in 2011. The same year, a video was created showing Rich Eisen, an NFL correspondent, side-by-side with Julio Jones running a 40-yard dash. A screenshot from the video is shown.

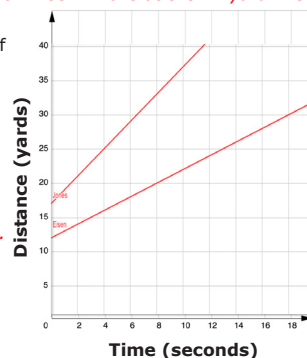


- a. What do you notice?

Answers will vary. Jones, at the 18 yard line, is running faster than Eisen who is at the 12 yard line.

- b. Make a sketch of what you think the graph of the two runners would look like. Be sure to label each line with the name of the runner.

Graphs will vary.



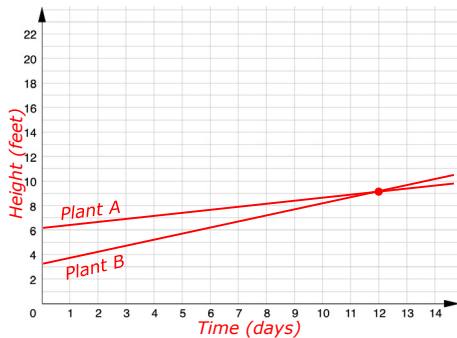


8.10.2 Guided Instruction: Solving Systems of Equations by Graphing in Real-World Contexts

Bamboo plants are known for growing very quickly. Imani measures the height of two bamboo plants over the course of several days. Plant A starts at 6 feet (ft) tall and grows at a constant rate of $\frac{1}{4}$ ft each day. Plant B starts at 3 ft tall and grows at a constant rate of $\frac{1}{2}$ ft each day.

1. The system $\begin{cases} y = \frac{1}{4}x + 6 \\ y = \frac{1}{2}x + 3 \end{cases}$ can be used to represent this situation, where x is the number of days and y is the height of the bamboo plant.

- a. Graph the system of equations.



- b. Label the axes with appropriate titles and units.
Shown on graph.
- c. Label each line Plant A or Plant B.
Shown on graph.

8.10.2 GUIDED INSTRUCTION: SOLVING SYSTEMS OF EQUATIONS BY GRAPHING IN REAL-WORLD CONTEXTS

DIFFERENTIATE INSTRUCTION

For question 1, model for students how to mark up the text while reading the question, including thinking aloud about why you're making those markings. It is important for students to hear the thought process for this strategy while also seeing it in action. Marking the text provides students the opportunity to make better sense of the context while also providing insight for the teacher about what information students are pulling from the text and what they may be missing.

TEACHER MOVES

Model how to complete question 1 for students as they actively take notes in their own workbooks. Consider asking students the following to revisit concepts from Unit 7:

- Which equation in the system represents each plant? Explain how you know using the slope and y-intercept.
- Which plant is growing at a faster rate?
- How would the graph change if the plants grew at the same rate?

8.10.2 GUIDED INSTRUCTION: SOLVING SYSTEMS OF EQUATIONS BY GRAPHING IN REAL-WORLD CONTEXTS (CONTINUED)

QUESTIONING

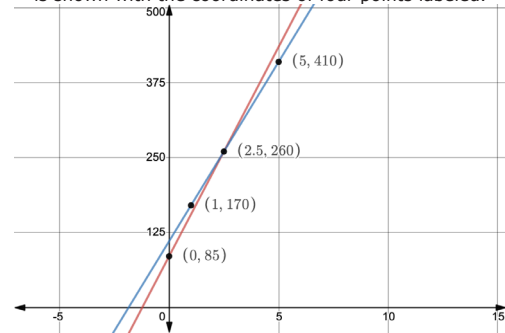
Consider having students verify the solution found graphically by using substitution.

TEACHER MOVES

Consider reading the situation in question 2 aloud while students mark the text on their own. Then, students could work in pairs to discuss and compare what information they marked and what they understand about the context before considering the solution and its meaning.

Listen as students discuss to determine student thinking to highlight and misconceptions to address when going through questions 2a and 2b with the whole class.

- d. Mark the solution to the system of equations on the graph. *Shown on graph.*
 - e. What are the coordinates of the solution?
(12, 9)
 - f. Explain the meaning of the solution in this context.
Plant A and Plant B will both be 9 feet tall after 12 days.
2. Sofia is deciding between two internet service providers. The first plan costs \$70 per month, with a one-time charge of \$85 for equipment. The second company charges \$60 per month, but the one-time equipment charge is \$110. The system $\begin{cases} y = 70x + 85 \\ y = 60x + 110 \end{cases}$ represents this situation, where x is the number of months and y is the total cost. The graph of the system is shown with the coordinates of four points labeled.



- a. What is the solution to the system?
(2.5, 260)
- b. Explain the meaning of the solution in this context.
Answers will vary. The total cost of both plans after 2.5 months will be \$260.

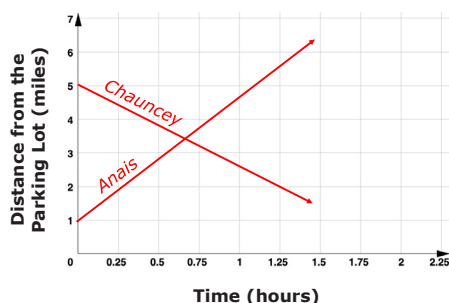


8.10.3 Guided Practice: Solving Systems of Equations by Graphing for Real-World Contexts

1. Chauncey and Anais both decide to head to the park. They hike from the parking lot to the lake and back, but they start their hikes at different times.

At the time that Chauncey reaches the lake and starts to turn back, Anais is 1 mile away from the parking lot and hiking at a constant speed of 3.75 miles per hour toward the lake. Anais' distance from the parking lot, d , can be expressed as $d = 3.75t + 1$, where t is the number of hours since Chauncey started walking back toward the parking lot.

Chauncey's distance from the parking lot, d , can be expressed as $d = -2.25t + 5$, where t represents the time, in hours, since he left the lake.



- a. Graph both lines on the coordinate plane.
Shown on graph.
- b. Label each line as "Anais" or "Chauncey."
Shown on graph.

- c. What is the approximate solution to the system $\begin{cases} d = 3.75t + 1 \\ d = -2.25t + 5 \end{cases}$?
Answers will vary. (0.7, 3.5)
- d. Explain the meaning of the solution in context.
Answers will vary. After 0.7 hours, Anais and Chauncey will both be 3.5 miles away from the parking lot.
- e. Explain why only the approximate solution for the system can be determined by graphing.
Answers will vary. Only an approximate solution for the system can be determined by graphing because the intersection is not on a grid line. Therefore, the solution to the system is an estimate.

8.10.3 GUIDED PRACTICE: SOLVING SYSTEMS OF EQUATIONS BY GRAPHING FOR REAL-WORLD CONTEXTS

DIFFERENTIATE INSTRUCTION

Allow informal observation throughout the first two activities to dictate how "guided" this Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

TEACHER MOVES

MA.K12.MTR.7.1: Real-World Contexts

Students apply their knowledge of solving systems of linear equations to solve real-world problems modeling two linear relationships. Students specifically use graphing to model and solve the real-world contexts problems, relating the mathematical solution to the real-world context it models.

DIFFERENTIATE INSTRUCTION

To enable students to verify the accuracy of their approximation, consider providing them with a graphing utility after completing question 1e.

ENRICHMENT

How does using a straightedge to graph the equations help to improve the accuracy when approximating the solution?

8.10.4 COLLABORATION: GRAPHING REAL-WORLD SYSTEMS

DIFFERENTIATE INSTRUCTION

Use observational data from 8.10.2 and 8.10.3 to form a small group of students who are struggling to work with systems of equations or interpreting them in context. Facilitate these students as they work through the 8.10.4 Collaboration.

TEACHER MOVES

Three Reads

Encourage students to use the “Three Reads” language routine to help comprehend the text. Have students read the text first for basic comprehension, second for analyzing the language and mathematical structure, and lastly to brainstorm possible solution paths.

TEACHER MOVES

Monitor while students are working to determine if there are any questions that need to be debriefed before the Wrap-Up. This could include highlighting student voices to share rich discussions or comments overheard while they collaborated for the whole class to hear, or to address common misconceptions that were evident among the majority of the class.



8.10.4 Collaboration: Graphing Real-World Systems

- Victor is monitoring the water levels in the control room at a water treatment plant. He receives an alert that Tank A has a leak. It started with 1,500 liters of water and is losing 2 liters of water per minute. At the same time as the leak began, Tank B held 300 liters of water and is being filled at a rate of 6 liters per minute.

This situation can be represented by the system $\begin{cases} y = -2x + 1500 \\ y = 6x + 300 \end{cases}$, where x is the amount of time, in minutes, since Victor received the alert and y is the amount of water, in liters, in the tank.

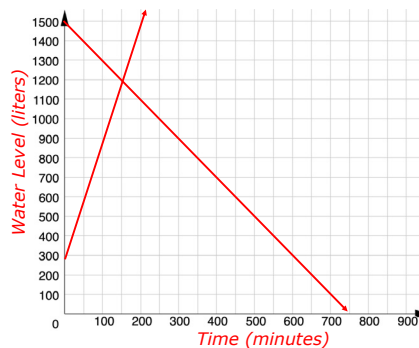
- Discuss with your partner which equation represents each tank. Then, complete the table.

Tank	Equation
Tank A	$y = -2x + 1500$
Tank B	$y = 6x + 300$

- Work with your partner to determine how much water will be in each tank 100 minutes after Victor received the alert.

Tank	Volume, in Liters, After 100 Minutes
A	1300 liters
B	900 liters

- On your own, graph the system of equations on the coordinate grid.



- Label both axes with appropriate titles and units.
- Compare your graph and labels with your partner and resolve any differences, if necessary.
- What is the solution to the system?
(150, 1200)
- Explain what the solution means in this context.
Answers will vary. 150 minutes after the alert, both tanks will contain 1200 liters of water.



8.10.5 Wrap-Up: Reflection

1. Take a **SELFIE** by checking the box that best matches what you did today.
Answers will vary.

Showed my work	☐	☐
Explained my answers	☐	☐
Lots of math vocabulary used	☐	☐
Found solutions by graphing	☐	☐
I persevered through the problem	☐	☐
Eliminated careless errors	☐	☐

8.10.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

QUESTIONING

What would reflection look like for you as a teacher now that the unit is completed? Consider your own scale for strengths and gaps for students through the lesson and how best to spiral solving systems of linear equation by graphing in future units.